M.M. Golam Hafiz

AI/ML Researcher | Medical Imaging | Computer Vision | Federated Learning | Trustworthy AI

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RESEARCH SUMMARY

AI/ML researcher focused on trustworthy and generalizable learning systems for medical imaging and distributed healthcare AI. Research spans uncertainty-aware image segmentation and classification, personalized and graph-based federated learning under heterogeneous non-IID data, and robustness across sites and domain shifts, with applications in gastrointestinal imaging, histopathology, and MRI. Additional work explores constraint-aware generative modeling, diffusion-based sensor translation, and reliable decision-making under uncertain structured knowledge, with an emphasis on rigorous ablation studies, cross-dataset validation, and reliability-aware evaluation.

EDUCATION

American International University-Bangladesh, Dhaka, Bangladesh 2022 – 2026
B.Sc. in Computer Science and Engineering
CGPA: 3.50/4.00

Hajera Taju Degree College, Chattogram, Bangladesh 2020
Higher Secondary Certificate (HSC)
GPA: 5.00/5.00

RESEARCH INTERESTS

Trustworthy Medical AI; Medical Image Analysis and Computational Pathology; Personalized and Federated Learning; Uncertainty Quantification, Calibration, and Selective Prediction; Domain Generalization and Cross-Site Robustness; Generative and Diffusion-Based Modeling; Graph-Based and Structured Learning

PUBLICATIONS AND MANUSCRIPTS

Journal Articles — Published

1. Muhammad Muhtasim Shahriar, **M.M. Golam Hafiz**, M. F. Mridha, and Md. Rakibul Islam. "Swin-MSEG-Pro: Uncertainty-aware boundary-focused Swin Transformer with deep supervision for reliable gastrointestinal polyp segmentation." *Array* (Elsevier), 2026. DOI: 10.1016/j.array.2026.101265
2. Muhammad Muhtasim Shahriar, **M.M. Golam Hafiz**, Md. Rakibul Islam, and M. F. Mridha. "FedPAXA-Net: Prototype-aligned personalized federated learning for multi-client gastrointestinal endoscopy image classification." *Intelligence-Based Medicine* (Elsevier), 2026, Article 100472. DOI: 10.1016/j.ibmed.2026.100472

Journal Manuscripts — Under Review or Submitted

3. Muhammad Muhtasim Shahriar, **M.M. Golam Hafiz**, Saad Aloteibi, and Mohammad Ali Moni. "FedHisto-PAST: Parameter-Efficient Stain-Aware Federated Learning for Cross-Site Lung Histopathology Classification." *Submitted to Engineering Applications of Artificial Intelligence (Elsevier).* arXiv: 2609.31150
4. **M.M. Golam Hafiz**, A. K. M. Shahriar Emtiaz, Safia Akter Simi, Md. Rakibul Islam, and M. F. Mridha. "Reliable Prerequisite-Aware Learning Path Recommendation under Uncertain Knowledge Structures: Long-Horizon Evaluation and Cross-Dataset Validation." *Under review at Computers & Education: Artificial Intelligence (Elsevier).*
5. **M.M. Golam Hafiz**, Safia Akter Simi, Md. Rakibul Islam, and M. F. Mridha. "DiffusionRef-IMU: Guide-Anchored Residual Diffusion for Cross-Location Inertial Measurement Unit Translation." *Under review at Results in Engineering (Elsevier).*
6. **M.M. Golam Hafiz**, Mahmud Al Ashiq, Pranto Biswas, Md. Rakibul Islam, and M. F. Mridha. "Isolating the Effects of VSS Depth Reduction and Residual Vision Mamba Refinement in VM-UNet for Medical Image Segmentation." *Submitted to Array (Elsevier).*
7. **M.M. Golam Hafiz**, Safia Akter Simi, Md. Rakibul Islam, and M. F. Mridha. "DiffBiz-Guard: Inference-Time Numerical Business-Rule Guidance for Auditable Synthetic Tabular Data Release" *Under review at Machine Learning with Applications (Elsevier).*

Conference Papers — Published

8. **M.M. Golam Hafiz**, Muhammad Muhtasim Shahriar, Md. Kishor Morol, Sadek Al Prince, Safia Akter Simi, Dip Nandi, and Md. Abdullah Al Jubair. "Federated Deep Learning with Client Communication Graphs for Osteosarcoma Histopathology Image Classification." *28th International Conference on Pattern Recognition (ICPR 2026)*, Lecture Notes in Computer Science, Vol. 16823, pp. 155–169, Springer, 2026. [Paper] [Code]
9. Muhammad Muhtasim Shahriar, **M.M. Golam Hafiz**, Riadul Islam Rabbi, Tze Hui Liew, Fatema Mostafa Tarin, and Ekramul Haque Tusher. "Comparative Analysis of Deep Learning Architectures for Brain Tumor Classification Using MRI Images." *2025 8th International Conference on New Media Studies (CONMEDIA)*, Malacca, Malaysia, 2025. [Paper] [Code]

Conference Papers — Accepted / To Appear

10. Md. Owafeeuzzaman Patwary, Apurbo Biswas, Muhammad Muhtasim Shahriar, **M.M. Golam Hafiz**, Morzia Momo, Kazi Tanvir, Md. Reazul Islam, and Dipta Gomes. "End-to-End Deep Learning Framework for Kidney MRI Segmentation and Explainable CKD Detection Using Transfer Learning." *In press, ECCT 2026.* [Conference]

11. **M.M. Golam Hafiz**, Md. Ashikur Rahman Shakib, Md. Maruf Hossain, Muntasir Mehrab, Md. Arafat Rahaman Sifat, and Abdus Salam. "GesturePolypSeg: An HCI Interface for Explainable and Uncertainty-Aware Binary Polyp Segmentation." *In press, ICCA 2026.* [Conference] [Code]

12. A. K. M. Shahriar Emtiaz, **M.M. Golam Hafiz**, Mamun Mahmud, Muhammad Muhtasim Shahriar, and Md. Kishor Morol. "Comparative Evaluation of CNN, Hybrid, and Transformer Architectures for Nuclei Segmentation in Histopathology." *In press, IEEE BECITHCON 2026.* [Conference]

13. Muhammad Muhtasim Shahriar, Mamun Mahmud, **M.M. Golam Hafiz**, Dipta Justin Gomes, and Md. Mohammad Ali Moni. "Privacy-Aware Federated Benchmarking and Explainability for Multi-Class Histopathology Image Classification." *In press, DICTA 2026, Sydney, Australia.* [Conference]

Manuscript in Preparation

14. **M.M. Golam Hafiz**, Safia Akter Simi, and M. F. Mridha. "Learning When to Verify: Verification-Gain Routing for Compute-Budgeted Cross-Architecture Polyp Segmentation under Domain Shift"

RESEARCH EXPERIENCE

AMIR Lab, American International University-Bangladesh, Dhaka, Bangladesh — February 2026 – Present

AI/ML Research Intern

Supervisor: Prof. Dr. Muhammad Firoz Mridha

Project 1: Swin-MSEG-Pro

Developed an uncertainty-calibrated, boundary-focused Swin Transformer with deep supervision for reliable gastrointestinal polyp segmentation, integrating segmentation performance with uncertainty-aware reliability analysis.

Project 2: FedPAXA-Net

Contributed to personalized federated learning for non-IID multi-client gastrointestinal endoscopy classification using client-local adaptation and reliability-weighted class-prototype aggregation under a controlled evaluation protocol.

Project 3: DiffBiz-Guard

Led the design of inference-time business-rule guidance for auditable synthetic tabular data, including guidance ablations, rule-validity analysis, and external validation. Drafted the first-author manuscript.

Project 4: VM-UNet

Led a controlled study isolating VSS-depth reduction and residual Vision Mamba refinement, with accuracy–efficiency comparisons and ablation analysis across three medical image segmentation datasets. Drafted the first-author manuscript.

Project 5: DiffusionRef-IMU

Developed guide-anchored residual diffusion for cross-location IMU translation, evaluating reconstruction fidelity, downstream activity recognition, and cross-dataset generalization.

Project 6: EviSelect-AV

Developing source-calibrated adaptive cross-architecture verification for reliable polyp segmentation under domain shift, with emphasis on uncertainty-aware verification and selective reliability.

Project 7: Prerequisite-Aware Learning Path Recommendation

Led reliable learning-path recommendation under uncertain prerequisite structures, including long-horizon evaluation, prerequisite perturbation analysis, and independent cross-dataset validation on MOOCCubeX and Junyi Academy data.

TECHNICAL SKILLS

- **Programming and Frameworks:** Python, PyTorch, TensorFlow, Keras, Hugging Face Transformers
- **Medical Imaging and Computer Vision:** Image Classification and Segmentation, CNNs, Vision Transformers, U-Net-Based Architectures, Transfer Learning, Multi-Scale Feature Fusion, Deep Supervision, Boundary-Aware Learning
- **Federated and Graph Learning:** FedAvg, Personalized Federated Learning, Non-IID Data and Client Heterogeneity Analysis, Client Communication Graphs, GraphSAGE, Topology-Aware Aggregation, Prototype-Guided Personalization
- **Uncertainty and Explainability:** Evidential Deep Learning, Predictive and Boundary Uncertainty, Selective Prediction Analysis, Grad-CAM, Feature-Importance Analysis
- **Feature Engineering and Experimental Evaluation:** Statistical and Spectral Feature Extraction, Prototype-Distance Features, Classical and Ensemble Learning, Ablation Studies, Cross-Dataset Evaluation, Paired Statistical Testing
- **Generative and Diffusion Modeling:** Diffusion Models, Conditional Diffusion, Flow Matching, Synthetic Tabular Data Generation, Constraint-Guided Generation

PROFESSIONAL EXPERIENCE

Agni Systems PLC, Dhaka, Bangladesh March 2026 – June 2026

Full Stack Web Developer — Full-time

Maintained agni.com using PHP, HTML, CSS, JavaScript, and Bootstrap; handled debugging, refactoring, and AI-generated code review for production integration.

Outlier AI, Remote September 2024 – January 2025

AI Trainer / AI Response Evaluator

Evaluated AI responses across text, image, location-based, and mathematical reasoning tasks for correctness, groundedness, instruction following, hallucinations, and omitted constraints.

REFERENCES

Prof. Dr. Muhammad Firoz Mridha
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